SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	**Dharshan B**	**Reg. No.:**	192312150
Section / Batch:	2023	**Date:**	26/10/26

Code of Conduct

I **Dharshan B 192312150** *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dharshan

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Engineer to Excel

MULTIPLE ACCESS TECHNIQUE : FDMA,TDMA,CDMA,OFDMA

ITA0328 - MOBILE COMPUTING

CO1 AT3 - INDIVIDUAL TALK

Guided By
Dr. VEERAMANI

Presented By
DHARSHAN B (192312150)

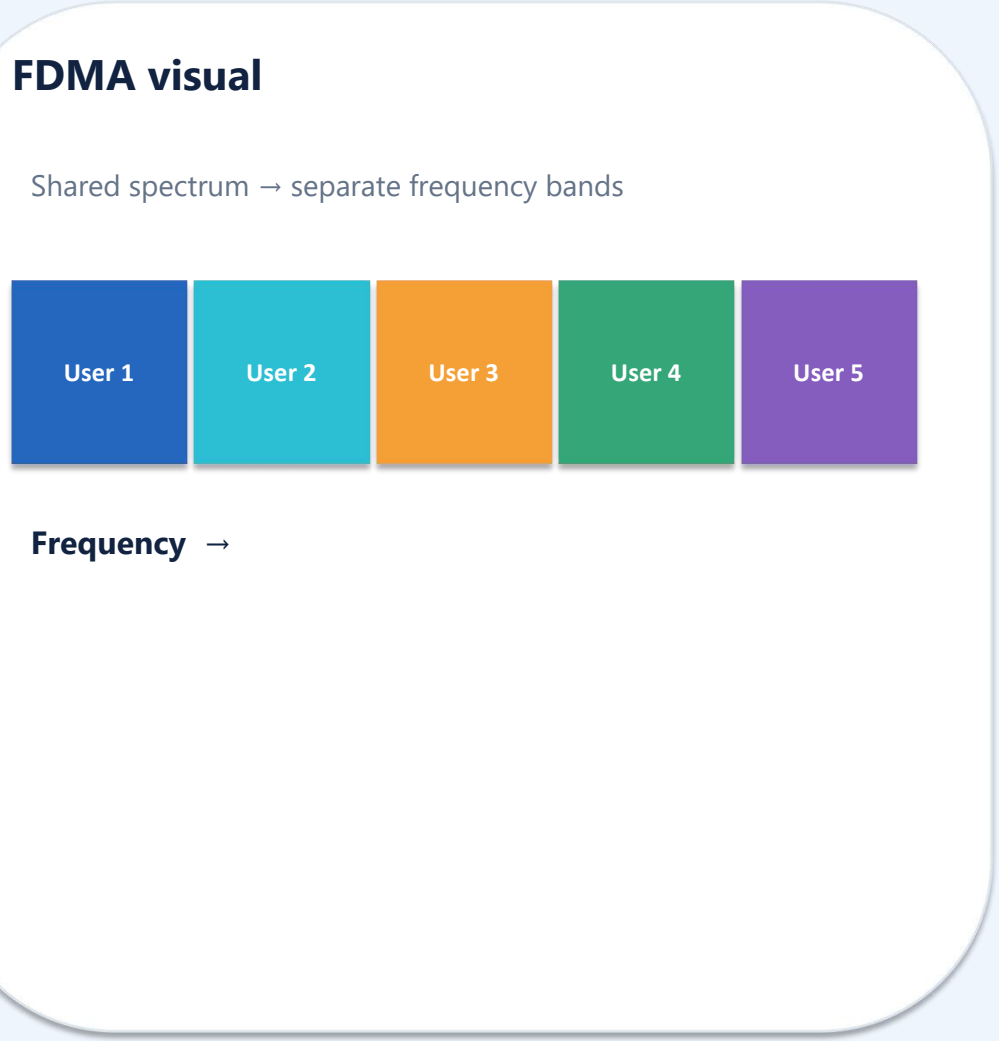
1. Introduction to Multiple Access

- Multiple access allows many users to share a communication medium.
- It improves the use of limited radio spectrum.
- Users are separated by frequency, time, code, or subcarriers.
- The right technique depends on capacity, complexity, and interference.
- These methods are used in cellular, satellite, and Wi-Fi systems.



2. FDMA — Frequency Division Multiple Access

- The total spectrum is divided into separate frequency channels.
- Each user communicates on an assigned frequency band.
- Several users can transmit at the same time on different frequencies.
- Guard bands help reduce interference between adjacent channels.
- Used in early 1G cellular systems and some satellite links.

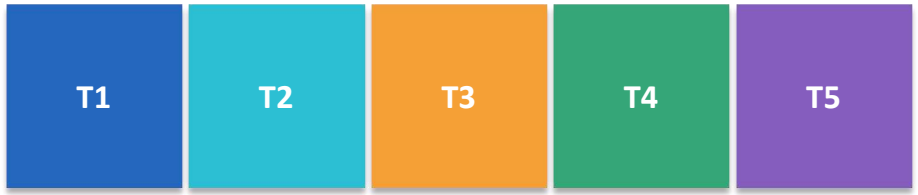


3. TDMA — Time Division Multiple Access

- A frequency channel is divided into repeating time slots.
- Each user transmits only during its assigned slot.
- Different users share the same frequency at different times.
- Accurate synchronization is needed to avoid slot overlap.
- Used in GSM and other digital communication systems.

TDMA visual

One frequency channel → time slots

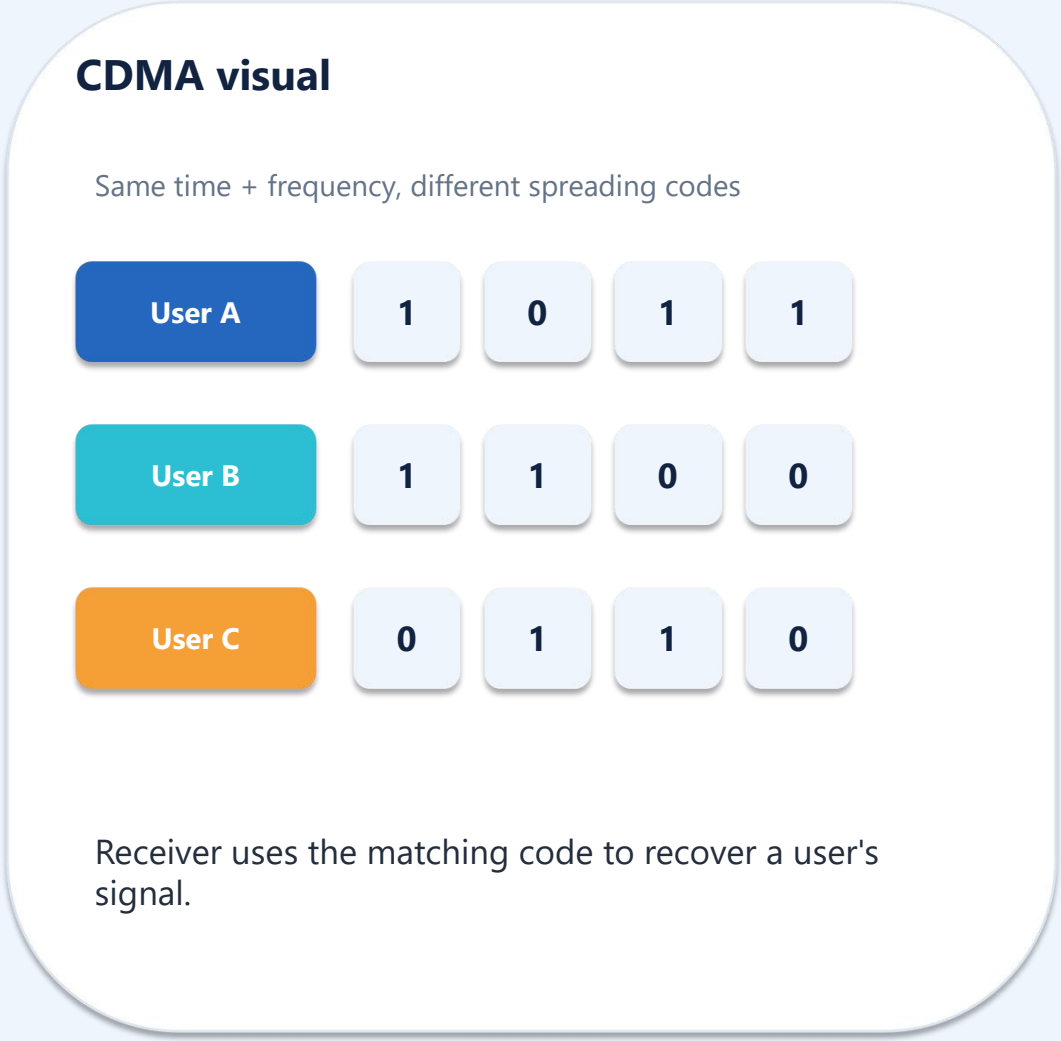


Time →

Each user transmits only during its assigned slot.

4. CDMA — Code Division Multiple Access

- All users can transmit over the same frequency band at the same time.
- Each user is assigned a unique spreading code.
- The receiver uses the correct code to identify the desired signal.
- It can resist some interference but needs complex signal processing.
- Used in IS-95, cdmaOne, and CDMA2000 systems.

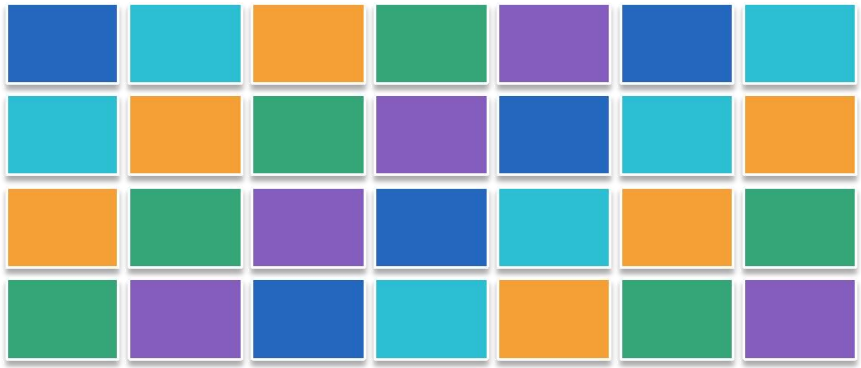


5. OFDMA — Orthogonal Frequency Division Multiple Access

- The channel is divided into many closely spaced orthogonal subcarriers.
- Different users receive different groups of subcarriers.
- It supports high data rates and flexible resource allocation.
- It handles multipath propagation well but needs good synchronization.
- Used in 4G LTE downlink, Wi-Fi 6, and 5G NR.

OFDMA visual

Time–frequency resource grid



Frequency ↑ **Time →**

Different users receive different groups of orthogonal subcarriers.

6. Comparison and Conclusion

Technique	User separation	Main strength	Main limitation	Typical examples
FDMA	Frequency bands	Simple concept	Guard bands needed	1G cellular; satellite links
TDMA	Time slots	Efficient time sharing	Needs synchronization	GSM
CDMA	Unique codes	Simultaneous shared access	Complex processing	IS-95; CDMA2000
OFDMA	Subcarrier groups	Flexible, high efficiency	Sensitive to synchronization errors	4G LTE downlink; Wi-Fi 6; 5G NR

Key takeaway

FDMA separates by frequency, TDMA by time, CDMA by code, and OFDMA by subcarriers. Together, these approaches help wireless networks share limited spectrum efficiently.


SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105
MULTIPLE ACCESS TECHNIQUE :
FDMA,TDMA,CDMA,OFDMA
ITA0328 - MOBILE COMPUTING
Presented By
DHARSHAN B (192312150)
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22h8+654, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.027891° Long 80.015524°
Monday, 26/10/2026 10:01 AM GMT +05:30
Note : Captured by GPS Map Camera
Google